(i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)=\frac{\pi}{4}$, since $\frac{\pi}{4}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$.

(ii) $\sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right)=-\frac{\pi}{3}$, since $-\frac{\pi}{3}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

(iii) $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)=\sin^{-1}\left(\sin\left(\pi-\frac{\pi}{6}\right)\right)=\sin^{-1}\left(\sin\frac{\pi}{6}\right)=\frac{\pi}{6}$, since $\frac{\pi}{6}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

Example 4.4

Find the domain of $\sin^{-1}\left(2-3x^2\right)$

Solution

We know that the domain of $\sin^{-1}(x)$ is $[-1,1]$.

This leads to $-1\le 2-3x^2\le 1$, which implies $-3\le -3x^2\le -1$.

Now, $-3\le -3x^2$, gives $x^2\le 1$ and ... (1)

$-3x^2\le -1$, gives $x^2\ge\frac{1}{3}$ ... (2)

Combining the equations (1) and (2), we get $\frac{1}{3}\le x^2\le 1$. That is, $\frac{1}{\sqrt{3}}\le|x|\le 1$, which gives

$x\in\left[-1,-\frac{1}{\sqrt{3}}\right]\cup\left[\frac{1}{\sqrt{3}},1\right]$, since $a\le|x|\le b$ implies $x\in[-b,-a]\cup[a,b]$.

EXERCISE 4.1

1. Find all the values of x such that

   (i) $-10\pi\le x\le 10\pi$ and $\sin x=0$ (ii) $-3\pi\le x\le 3\pi$ and $\sin x=-1$.

2. Find the period and amplitude of

   (i) $y=\sin 7x$ (ii) $y=-\sin\left(\frac{1}{3}x\right)$ (iii) $y=4\sin(-2x)$.

3. Sketch the graph of $y=\sin\left(\frac{1}{3}x\right)$ for $0\le x<6\pi$.

4. Find the value of (i) $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$ (ii) $\sin^{-1}\left(\sin\left(\frac{5\pi}{4}\right)\right)$.

5. For what value of x does $\sin x=\sin^{-1}x$?

6. Find the domain of the following

   (i) $f(x)=\sin^{-1}\left(\frac{x^2+1}{2x}\right)$ (ii) $g(x)=2\sin^{-1}(2x-1)-\frac{\pi}{4}$.

7. Find the value of $\sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9}+\cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$.

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